

Topic 3 Summary: Relative Motion

- Definitions
- Relative motion in 1D and 2D
- Graphical solution (using law of sines and law of cosines)
- Solution by writing the vectors in Cartesian (component) form
- Worked examples

RELATIVE MOTION

Strictly speaking, all motion is relative to something.

Usually that something is a reference point that is assumed to be at rest (i.e. the earth).

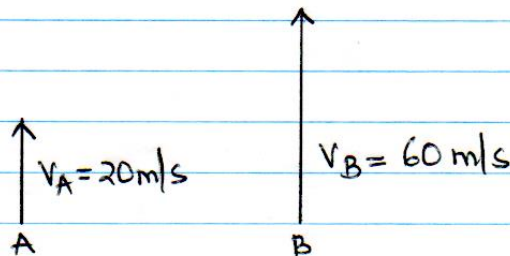
Motion can be relative to anything, even another moving object

Consider two aeroplanes A and B flying at the same height. A is flying 200 km/h due North and B at 400 km/h on a course 030° . These are their velocities relative to the ground and are their true velocities. We denote velocity of A, V_A and that of B, V_B .

To the pilot of A, however, B will seem to have a different velocity because he himself is moving. We call this apparent velocity, the velocity of B relative to A and denote it, V_{BA} . Similarly B will see A moving with the velocity of A relative to B, V_{AB} .

ONE - DIMENSIONAL CASE:

Suppose car A is moving with 20 m/s due North and car B is moving at 60 m/s due North. i.e.



Then the velocity of B w.r.t. A (B relative to A) is

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$

$$= 60\hat{j} - 20\hat{j} = 40\hat{j} \text{ m/s.}$$

Similarly, velocity of A relative to B,

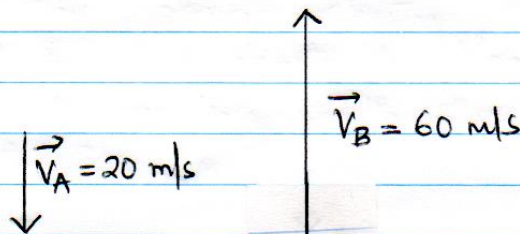
$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$$

$$= 20\hat{j} - 60\hat{j} = -40\hat{j} \text{ m/s}$$

$$\text{i.e. } \vec{V}_{BA} = -\vec{V}_{AB}$$

⇒ The magnitude of the relative velocity between car A and car B is the same.

Now suppose A and B were moving in opposite directions as shown.



$$\text{Then } \vec{V}_{BA} = \vec{V}_B - \vec{V}_A = 60\hat{j} - (-20)\hat{j} = 80\hat{j} \text{ m/s}$$

$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B = -20\hat{j} - 60\hat{j} = -80\hat{j} \text{ m/s.}$$

Note:- The velocity of A relative to B can also be described as the velocity which "A appears to have" to an observer moving with B.

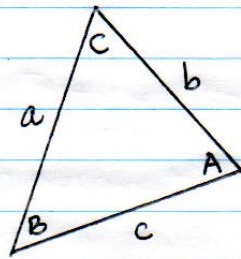
$$V_{\text{relative}} = V_{\text{object}} - V_{\text{observer}}$$

Relative velocity problems involve the addition and subtraction of vectors whether in 1D or 2D.

These problems can be solved in one of two ways.

a) Graphically

This approach involves drawing a sketch and then using trigonometry (law of sines and law of cosines) to solve.



$$\text{law of sines: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

b) by writing the vectors in Cartesian form (component form) i.e. $\vec{v} = a\hat{i} + b\hat{j}$.

Note: If a triangle of vectors is drawn, each side must be fully labelled with the magnitude of the vectors in brackets.

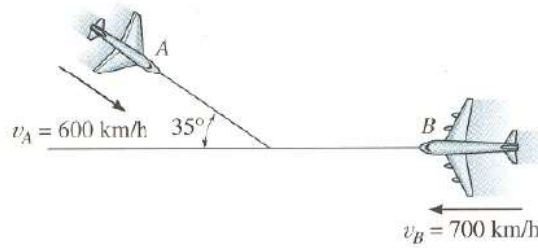
Displacements and acceleration are relative too.

The absolute position of two particles A and B w.r.t. x-y-z reference frame are given by $\vec{r}_A$ and $\vec{r}_B$.

Then
$$\vec{r}_{AB} = \vec{r}_A - \vec{r}_B$$

Accelerations are relative similarly.

EXAMPLE



Given: $v_A = 600 \text{ km/hr}$
 $v_B = 700 \text{ km/hr}$
Find: $v_{B/A}$

Plan:

- Vector Method: Write vectors $\mathbf{v}_A$ and $\mathbf{v}_B$ in Cartesian form, then determine $\mathbf{v}_B - \mathbf{v}_A$
- Graphical Method: Draw vectors $\mathbf{v}_A$ and $\mathbf{v}_B$ from a common point. Apply the laws of sines and cosines to determine $v_{B/A}$.

Solution:

- Vector Method:

$$\begin{aligned}\mathbf{v}_A &= 600 \cos 35^\circ \mathbf{i} - 600 \sin 35^\circ \mathbf{j} \\ &= (491.5 \mathbf{i} - 344.1 \mathbf{j}) \text{ km/hr}\end{aligned}$$

$$\mathbf{v}_B = -700 \mathbf{i} \text{ km/hr}$$

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (-1191.5 \mathbf{i} + 344.1 \mathbf{j}) \text{ km/hr}$$

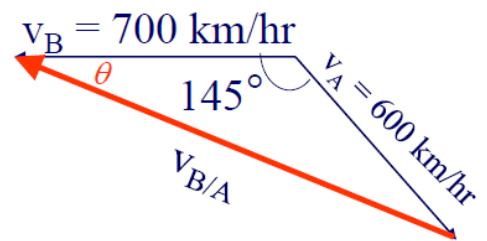
$$v_{B/A} = \sqrt{(1191.5)^2 + (344.1)^2} = 1240.2 \frac{\text{km}}{\text{hr}}$$

$$\text{where } \theta = \tan^{-1}\left(\frac{344.1}{1191.5}\right) = 16.1^\circ$$



b) **Graphical Method:**

Note that the vector that measures the tip of B relative to A is $\mathbf{v}_{B/A}$.



Law of Cosines:

$$v_{B/A}^2 = (700)^2 + (600)^2 - 2(700)(600)\cos 145^\circ$$

$$v_{B/A} = 1240.2 \frac{\text{km}}{\text{hr}}$$

Law of Sines:

$$\frac{v_{B/A}}{\sin(145^\circ)} = \frac{v_A}{\sin\theta} \quad \text{or} \quad \theta = 16.1^\circ$$

Example 2.

To an observer in a train travelling due East at 40 km/h, an aeroplane appears to be travelling due North at 75 km/h. Find the true course and speed of the aeroplane.

Solution

let V_A - Velocity of aeroplane
 V_T - Velocity of train

$$V_{AT} = V_A - V_T$$

$$\therefore V_A = V_{AT} + V_T$$

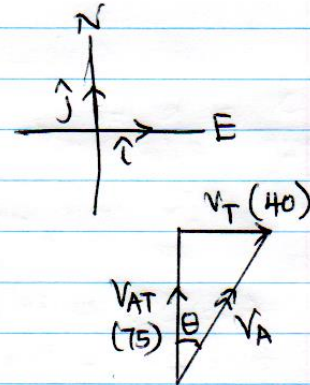
$$\text{Given } V_{AT} = 75\hat{j}$$

$$V_T = 40\hat{i}$$

$$\therefore V_A = 75\hat{j} + 40\hat{i}$$

$$|V_A| = \sqrt{75^2 + 40^2} = 85 \text{ km/h.}$$

$$\tan \theta = \frac{40}{75} \Rightarrow \alpha = 28.1^\circ \text{ East of North.}$$



Example 3:

A man running due North at 8 m/s experiences a wind which appears to come from the North-East at 10 m/s. Find the true velocity of the wind.

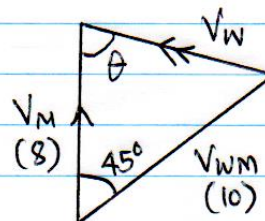
Solution

$$V_M = 8 \text{ m/s N}$$

$$V_{WM} = 10 \text{ m/s NE}$$

$$V_{WM} = V_W - V_M$$

$$\therefore V_W = V_{WM} + V_M$$



Using cosine rule

$$|V_w|^2 = 8^2 + 10^2 - 2(8)(10)\cos 45^\circ$$
$$= 50.8$$

$$|V_w| = 7.132 \text{ m/s.}$$

Using sine law

$$\frac{\sin \theta}{10} = \frac{\sin 45^\circ}{7.132} \Rightarrow \theta = 82.52^\circ$$

$\therefore$ Velocity of wind is 7.132 m/s in the direction 82.52° West of North.

Ex. Attempt example 3 using components of vectors.

Let $\hat{i}$ and $\hat{j}$ be directed towards East and North respectively.

Example 4:

A light plane attains an airspeed of 500 km/h. The pilot sets out for a destination 800 km to the North but discovers that the plane must be headed 20° E of N to fly there directly. The plane arrives in 2 hrs. What wills the wind velocity vector?

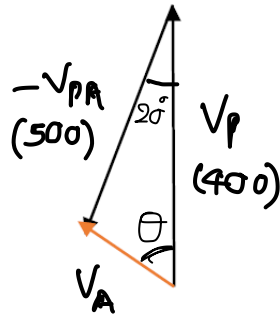
Solution

$$V_{PA} = \text{Velocity of plane w.r.t air (airspeed)} = 500 \text{ km/h.}$$

$$V_P = \frac{800}{2} = 400 \text{ km/h}$$

$$V_{PA} = V_P - V_A$$

$$\therefore V_A = V_P - V_{PA}$$



$$|V_A| = 400^2 + 500^2 - 2(400)(500)\cos 20$$

$$= 184.7 \text{ km/h.}$$

$$\frac{\sin \theta}{500} = \frac{\sin 20}{184.7} \Rightarrow \theta = 67.8^\circ$$

a direction

The wind is blowing at 184.7 km/h ~~in~~ from 67.8° W of N.

Example 5:

Ship P is initially 500m due West of ship Q. Given that P sails NE at 15 m/s and Q sails due North at 8 m/s, find:

- the velocity of P relative to Q
- the shortest distance between them.

Solution

$$\begin{aligned} \text{a) } V_P &= 15 \cos 45 \hat{i} + 15 \sin 45 \hat{j} \\ &= 10.6 \hat{i} + 10.6 \hat{j} \\ V_Q &= 8 \hat{j} \end{aligned}$$

$$\begin{aligned} \therefore V_{PQ} &= V_P - V_Q \\ &= \begin{pmatrix} 10.6 \\ 10.6 \end{pmatrix} - \begin{pmatrix} 0 \\ 8 \end{pmatrix} = \begin{pmatrix} 10.6 \\ 2.6 \end{pmatrix} \Rightarrow |V_{PQ}| = \sqrt{10.6^2 + 2.6^2} \\ &= 10.92 \text{ m/s} \end{aligned}$$

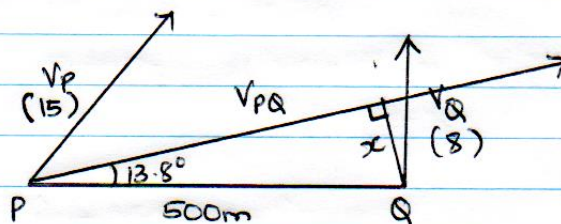
$$\tan \theta = \frac{2.6}{10.6} \Rightarrow \theta = 13.8^\circ$$

$\therefore V_{PQ}$ is 10.92 m/s at 076.2° (bearing).

- b) The only way to work out the shortest distance in subsequent motion between two moving objects is to make one of the objects seem to be stationary. Then analyse the motion of second object relative to this first one.

Since we have already worked out V_{PQ} , we have assumed that Q is the stationary ship.

So now we draw a diagram and indicate on it ship Q and initial position of ship P, along with the relative velocity V_{PQ} . i.e.



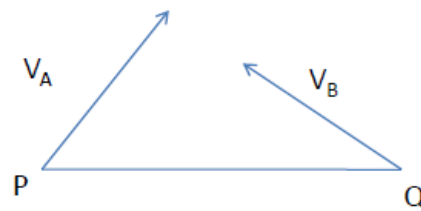
The shortest distance (x) is the perpendicular distance from Q to the line of V_{PQ} .

$$x = 500 \sin 13.8 = \underline{\underline{119.1 \text{ m}}}$$

Displacement is relative too!!

Consider two ships A and B initially at points P and Q moving with constant velocities V_A and V_B respectively. We can then find the V_{AB} . This relative velocity considers A moving while B is taken to be stationary. Thus if V_{AB} is said to be in the direction PQ then most likely A will get to Q and collide with B.

Diagram;

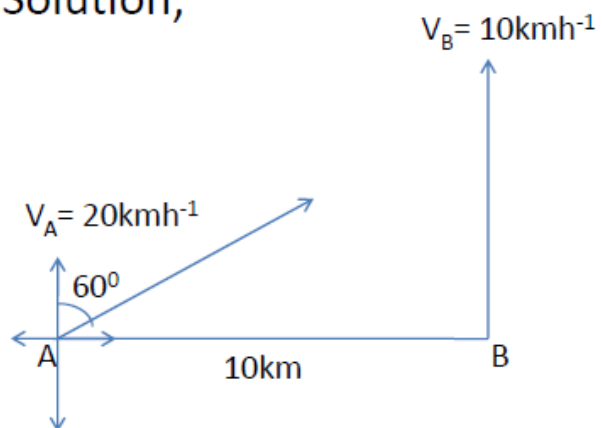


When the direction of motion for A and B is known and V_{AB} calculated to be in the direction from P to Q then collision of A and B is expected. When we use vectors, the vectors V_{AB} and r_{AB} are in the same direction and thus parallel.
 $V_{AB} = V_A - V_B$ and $r_{AB} = r_A - r_B$.

Example

At 12 noon two ships A and B are 10km apart with B due east of A. A is travelling at 20kmh^{-1} in a direction $N 60^\circ E$ and B is travelling at 10kmh^{-1} due north. Show that, if the two ships maintain these velocities, they will collide and find, to the nearest minute, when the collision occurs.

Solution;



$$V_A = 20\cos 30^\circ \mathbf{i} + 20\sin 30^\circ \mathbf{j}.$$

$$V_A = 10\sqrt{3}\mathbf{i} + 10\mathbf{j}.$$

$$V_B = 10\mathbf{j}.$$

If we assume A to be at the origin
then B will be at (10,0).

Hence $r_A = 0i + 0j$

$$r_B = 10i.$$

$$r_{AB} = r_A - r_B = (0i + 0j) - 10i \\ = -10i.$$

$$V_{AB} = V_A - V_B \\ = 10\sqrt{3}i + 10j - 10j \\ = 10\sqrt{3}i$$

Since $r_{AB} = KV_{AB}$

Then the relative velocity V_{AB} is
in the direction of the relative
displacement r_{AB} . Hence
collision is expected.

Time = Distance/velocity

$$T = \frac{|r_{AB}|}{|V_{AB}|}$$

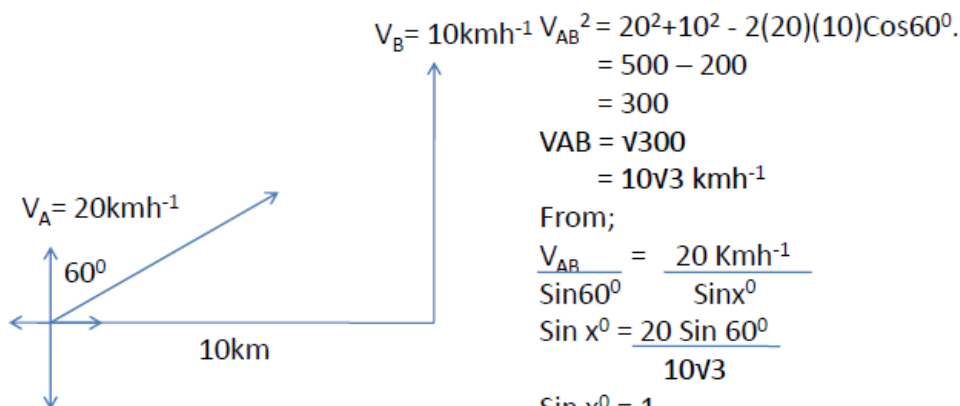
$$T = \frac{\sqrt{(-10)^2}}{\sqrt{(10\sqrt{3})^2}}$$

$$= 0.57735 \text{ Hrs}$$

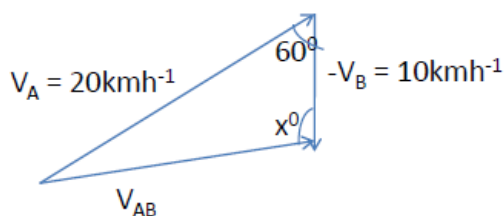
$$= 34.6 \text{ minutes} = 35 \text{ minutes.}$$

Therefore collision occurs at **12.35p.m**

Using the sketch



To find $V_{AB} = V_A - V_B$
 $= V_A + -V_B$



$$\sin x^\circ = 1$$

$$x^\circ = \sin^{-1}(1)$$

$$x^\circ = 90^\circ.$$

Direction of V_{AB} is due east hence collision is expected.